

Orr (Orri) Zussman

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EDUCATION

Columbia University, School of Engineering & Applied Science

Expected May 2028

B.S. in Mechanical Engineering, Minors in Aerospace & Jewish Studies

- **Relevant Coursework:** Computer Graphics & Design, Intro to Applied Math, Foundations of Data Science, Mechanics | *Fall 2026:* Mechanics of Solids, Mechanics of Fluids, Thermodynamics, Mechanical Eng. Lab I, Intro to Machining

RELEVANT EXPERIENCE

Columbia Creative Machines Lab (Prof. Hod Lipson), New York, NY

Feb. 2026 – Present

Research Project Lead (May 2026 – Present), Research Assistant (Feb. 2026 – May 2026)

- Own design process of an actuating treadmill rendering variable terrain at 100 mm height range and 25 × 14 mm segment resolution, directing a 4-person team.
- Reduced servo count by 97% (from 720 to 20) by redesigning the terrain to be “printed” rather than fully actuated.
- Expanding team, designing next-generation per-segment leadscrew architecture to increase resolution.

FireDome: Fire Detection & Suppression, Tel Aviv, Israel (Hybrid)

Mar. 2025 – Aug. 2025

Engineering Intern, Interim Job Site Manager (June 2025 – Aug. 2025)

- Designed a valve assembly in Fusion 360 for a 30 L latex balloon, integrating the detonation mechanism and packaging for an automatic fill-and-load system.
- Engineered a cost-neutral redesign of the high-voltage capacitor banks, cutting assembly man-hours by 50%.
- Assembled an 8-ton launcher system with a 4-person crew over 4 weeks (SMAW welding, magnetic drill press), executing field modifications at height to resolve post-production design conflicts.
- Directed heavy equipment (boom crane, manlifts, loaders) and oversaw site safety as Interim Job Site Manager.

Procurement Intern (Mar. 2025 – June 2025)

- Sourced US-based suppliers for launcher components, enabling domestic assembly and manufacturing.

Columbia Robotics Club, New York, NY

Dec. 2025 – Present

Columbia Combat Robotics, Engineering Lead

- Waterjet (steel, aluminum, titanium), formed sheet metal, and 3D printed engineering materials (PA6-GF, TPU) for Roar-E, a 12 lb combat robot that qualified for the NHRL September Pro Tour; designing 30lb for Feb. competition.
- Executed repairs and turnarounds in the pit, restoring robot to competition readiness under event time constraints.

EXTRACURRICULARS

American Society of Mechanical Engineers (ASME), New York, NY

Sept. 2026 – Present

Student Representative, Columbia Chapter

Columbia Space Initiative (CSI), New York, NY

Sept. 2024 – Present

Rockets Team (Sept. 2026 – Present), NASA Lunabotics Challenge (Sept. 2024 – Dec. 2024)

- Member of the propulsion fluids subteam for CSI’s first hybrid rocket.
- Worked on the drill of Columbia’s moon rover for NASA’s Lunabotics Challenge.

Manhattan High School President’s Council (MHSPC), New York, NY

Oct. 2022 – Apr. 2025

NYC Department of Education (DOE) Manhattan High Schools District Student Representative

- Represented 56,000 students at the DOE, including for the superintendent hiring committee.

AWARDS & HONORS

Special Congressional Recognition, Rep. Adriano Espaillat (2025); Certificate of Recognition, NY State Sen. Cordell Cleare (2025); Presidential Volunteer Service Award, Gold Medal (2023); New York State Comptroller Award, Comptroller Thomas DiNapoli (2023); Presidential Volunteer Service Award, Gold Medal (2022)

SKILLS

Manufacturing: Waterjet cutting, laser cutting, sheet metal forming, SMAW welding, soldering, 3D printing (FDM & SLA), power tools (magnetic drill press, band saw, rotary hammer, angle grinder), eng. drawing, GD&T, CNC Mill & Lathe
Software: SolidWorks, Fusion 360, Inventor, Python, NI Multisim | **Languages:** English & Hebrew (native)